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Exp 3: Study of network cables and crimping with RJ45 connector.

Aim:

To study different types of network cables and to prepare a straight-through and cross-over Ethernet cable by crimping RJ45 connector

Procedure:

1) Cut the required length of UTP cable using cable cutters.

2) Strip approximately 2.5 cm of outer jacket using a cable stripper.

3) Untwist the wire pairs and arrange them according to TIA/EIA 568B order.

4) Trim all wires to equal length (1.5 cm from the jacket).

5) Insert the wires into the RJ45 connector, ensuring each wire reaches the front

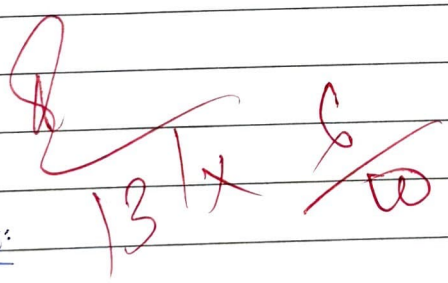
6) Crimp firmly using the crimping tool.

7) Repeat the same wiring sheme on the other end for a straight-through cable.

```
netstat -tuln              # Linux - Listening ports
nslookup google.com        # DNS resolution
arp -a                     # ARP table
route -n                   # Routing table (Linux)
```

8) Test the crimped cable using a cable tester to verify continuity.

9) Record the LED sequence observed during testing.



Result:

Both straight-through and cross-over Ethernet cables were successfully crimped and verified using a cable tester. All 8 pins showed proper continuity.